

GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - DTC: AUTO
HIGH BEAM CONTROL MODULE [AHBCM] [G1452021]

DESCRIPTION AND OPERATION

AUTO HIGH BEAM CONTROL MODULE [AHBCM]

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check JLR claims submission system for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Auto High Beam Control Module (AHBCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual. For additional information, refer to: [Headlamps](#) (417-01 Exterior Lighting, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1286-16	Interior mirror - Circuit voltage below threshold	- Auto high beam control module power or ground circuit open circuit, high resistance - Battery/charging system fault - Auto high beam control module internal failure	NOTE: The auto high beam control module will self recover after the supply voltage reaches normal conditions - Using the manufacturer approved diagnostic system, check the datalogger signal - main ECU supply voltage (0xDD02) - Refer to the electrical circuit diagrams and check the auto high beam control module power and ground circuit for open circuit, high resistance - Check the voltage between pin 1 and pin 3 of the auto high beam control module connector. The voltage should to be above 8.5 volts for normal conditions. Using the manufacturer approved diagnostic system, clear the DTC and retest - Refer to the relevant section of the workshop manual and test the battery and charging system - If the DTC is set permanent and the voltage is above 8.5 volts, check and install a new auto high beam control module as required
B1286-17	Interior mirror - Circuit voltage above threshold	- High voltage detected at auto high beam control module (battery voltage > 16.5 volts) - Battery/charging system fault - Auto high beam control module internal failure	NOTE: The auto high beam control module will self recover after the supply voltage reaches normal conditions - Using the manufacturer approved diagnostic system, check the datalogger signal - main ECU supply voltage (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system - Check the voltage between pin 1 and pin 3 of the auto high beam control module. The voltage should to be below 16.5 volts for normal conditions. Using the manufacturer approved diagnostic system, clear the DTC and retest - If the DTC is set permanent and voltage is below 16.5 volts, check and install a new auto high beam control module as required
B1286-44	Interior mirror - Data memory failure	Auto high beam control module - Internal failure	Clear DTC and retest. If fault persists, check and install a new auto high beam control module as required
B1286-47	Interior mirror - Watchdog/safety micro controller failure	Auto high beam control module - Internal failure	Clear DTC and retest. If fault persists, check and install a new auto high beam control module as required
B1286-49	Interior mirror - Internal electronic failure	Auto high beam control module - Internal failure	Clear DTC and retest. If fault persists, check and install a new auto high beam control module as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1286-60	Interior mirror – Reserved by document	Auto high beam control module - Operation temperature below limit	Allow the vehicle interior temperature to increase, the auto high beam control module will self recover after temperature reaches normal conditions
B1286-78	Interior mirror - Alignment or adjustment incorrect	Auto high beam control module low sensitivity mode has been activated	- The low sensitivity mode is cancelled when the ignition is cycled - Clear the DTC and retest
B1286-96	Interior mirror - Component internal failure	- Lost communication with the central junction box - Auto high beam control module - Internal failure	- Using the manufacturer approved diagnostic system, complete a CAN network integrity test - Refer to the electrical circuit diagrams and check the CAN network between the central junction box and the auto high beam control module - Using the manufacturer approved diagnostic system, clear the DTC and retest. If the fault persists, install a new auto high beam control module
B1286-97	Interior mirror - Component or system operation obstructed or blocked	- Auto high beam control module temporary disabled - Auto high beam control module camera has detected a partial or total blockage of the field of view	- Check to see if anything is blocking the field of view of the auto high beam control module camera - Using the manufacturer approved diagnostic system, clear DTC and retest - The auto high beam control module will self recover after failure condition has been rectified
B1286-98	Interior mirror - Component or system over temperature	Auto high beam control module temporary disabled because of operation temperature above limit	Auto high beam control module will self recover after the temperature reaches normal condition
B12AC-11	Electrochromic Door Mirror - Circuit short to ground	Electrochromic door mirror output circuit - Short circuit to ground	NOTE: To aid fault finding use the manufacturer approved diagnostic system to force the automatic dim function Refer to the electrical circuit diagrams and check the auto high beam control module harness, including both exterior mirror circuits for short circuit to ground, repair as necessary
B12AC-12	Electrochromic Door Mirror - Circuit short to battery	Electrochromic door mirror output circuit - Short circuit to power	NOTE: To aid fault finding use the manufacturer approved diagnostic system to force the automatic dim function Refer to the electrical circuit diagrams and check the auto high beam control module harness, including both exterior mirror circuits for short circuit to ground, repair as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B12EB-78	Camera horizontal alignment - Alignment or adjustment incorrect	- Auto high beam control module internal camera misaligned in the horizontal direction - Auto high beam control module fault - Windscreen alignment incorrect	NOTE: To trigger the possible fault conditions for this DTC, the vehicle may need to be driven during night time conditions - Check the auto high beam control module for security and correct positioning. Clear the DTC and retest for normal operation - Renew the auto high beam control module. See topix workshop manual, section 417-01 exterior lighting, description and operation for the auto aim calibration procedure - If the problem persists, suspected windscreen alignment incorrect
B12EC-78	Camera vertical alignment - Alignment or adjustment incorrect	- Auto high beam control module internal camera misaligned in the vertical direction - Auto high beam control module fault - Windscreen alignment incorrect	NOTE: To trigger the possible fault conditions for this DTC, the vehicle may need to be driven during night time conditions - Check the auto high beam control module for security and correct positioning. Clear the DTC and retest for normal operation - Renew the auto high beam control module. See topix workshop manual, section 417-01 exterior lighting, description and operation for auto aim calibration procedure - If the problem persists, suspected windscreen alignment incorrect
B134A-78	Target aim verification - Camera horizontal alignment - Alignment or adjustment incorrect	NOTE: This DTC is for use during the factory aim routine only - Auto high beam control module camera misaligned in the horizontal direction. The windscreen button and the auto high beam control module attachment alignment incorrect - Auto high beam control module internal failure	- Check accuracy of mirror attachment and windscreen button - Check and install new auto high beam control module as required
B134B-78	Target aim vertical - Camera horizontal alignment - Alignment or adjustment incorrect	NOTE: This DTC is for use during the factory aim routine only - Auto high beam control module camera misaligned in the vertical direction. The windscreen button and the auto high beam control module attachment alignment incorrect - Auto high beam control module internal failure	- Check accuracy of mirror attachment and windscreen button - Check and install new auto high beam control module as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0010-88	Medium speed CAN communication bus - Bus off	Medium speed CAN bus circuit fault	- Refer to the electrical circuit diagrams and check the power and ground connections to the auto high beam control module - Using the manufacturer approved diagnostic system, complete a CAN network integrity test - Refer to the electrical circuit diagrams and check the CAN network
U0140-00	Lost communication with Body Control Module (BCM) - No sub type information	Lost communication with central junction box	- Refer to the electrical circuit diagrams and check the power and ground connections to the auto high beam control module - Using the manufacturer approved diagnostic system, complete a CAN network integrity test - Refer to the electrical circuit diagrams and check the CAN network between the auto high beam control module and central junction box
U0300-00	Internal control module software incompatibility - No sub type information	Auto high beam control module software incompatibility	Using the manufacturer approved diagnostic system, ensure that the module contains the latest software version and is correctly configured, update if necessary. Clear the DTC and retest
U201A-57	Control module main calibration data - Invalid/incomplete software component	- Incorrect auto high beam control module fitted - Main calibration is invalid to car configuration file or not complete stored to the auto high beam control module	NOTE: Due to the mechanical calibration tolerance the correct auto high beam control module assembly must be used for the vehicle model types in question and it is not exchangeable with other vehicle model types - Using the manufacturers approved diagnostic system, compare the vehicle type in the auto high beam control module with the car configuration file (parameter 1), if these do not match the incorrect auto high beam control module has been fitted - Using the manufacturers approved diagnostic system, check the configuration of the car configuration file and the software version of the auto high beam control module is correct
U2100-00	Initial configuration not complete - No sub type information	- Invalid/incomplete software component - Auto high beam control module internal failure	- Using the manufacturers approved diagnostic system, check the configuration of the car configuration file and software version of the module is correct - Using the manufacturer approved diagnostic system check the central junction box for related DTCs and refer to the relevant DTC index - If the problem persists, renew the auto high beam control module
U2101-00	Control module configuration incompatible - No sub type information	Auto high beam control module is not configured correctly	Using the manufacturer approved diagnostic system check and update the car configuration file as required. Clear the DTC and retest